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Yearly energy performance of a photovoltaic-phase change material (PV-PCM) system in hot climate

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ABSTRACT

A photovoltaic-phase change material (PV-PCM) system is employed in extremely hot environment of the United Arab Emirates (UAE) to evaluate its energy saving performance throughout the year. A paraffin based PCM with melting range of 38–43 °C is integrated at the back of the PV panel and its cooling effect is monitored. The increased PV power output due to cooling produced by PCM is quantified. A Conjugate heat transfer model employing enthalpy based formulation is developed and validated with the experimental data. The model is employed to predict melting and solidification fractions in each month of the year. The PV-PCM is found to exhibit consistent performance for most of the year. The PCM produced less cooling in peak cool and peak hot months attributed to its incomplete melting and solidification, respectively. The PV-PCM system enhanced the PV annual electrical energy yield by 5.9% in the hot climatic conditions.

1. Introduction

Various active and passive approaches have been adopted to mitigate the adverse effects of photovoltaics (PV) temperature on their performance reviewed in (Du et al., 2013; Hasanuzzaman et al., 2016). The passive approaches achieved temperature drop up to 20 °C with an improvement in electrical efficiency up to 15.5%. The passive approaches are deemed least effective in cooling PV in warmer climates where PV temperature (T_{PV}) rises frequently above 70 °C. Active approaches can effectively cool PV up to 30 °C and improve the electrical efficiency up to 22% (Hasanuzzaman et al., 2016) however, these consume additional energy to keep the coolant flow thus surpassing the energy saved by PV cooling (Du et al., 2013). The merits and demerit of the conventional PV cooling approaches are summarized in Table 1 (Hasan et al., 2010) which identifies a relatively efficient phase change material (PCM) based PV cooling technique (Huang et al., 2004; Hasan et al., 2010).

The PCM based passive PV cooling approach have recently gained more attention from PV community. The research into PCMs characteristics and selection criteria for effective PV cooling (Hasan et al., 2014a) concluded that the selection of optimum PCM in varying weather conditions needs deeper understanding of PCM performance. PCM containment materials (Huang et al., 2006), PCM combinations, metallic conductivity enhancing fins (Huang, 2011) and solar radiation intensities (Hasan et al., 2010) have been reported to influence performance of the photovoltaic-phase change material (PV-PCM) systems to greater extent. The suggested thermal conductivity enhancing fins have been reported to cool PV effectively under single sun (Hasan et al., 2014a, 2015; Kibria et al., 2016) and at low solar radiation concentration (Sharma et al., 2016; Ceylan et al., 2016). Although, integration of PCM into the PV systems can achieve lower PV temperatures, it incurs additional capital cost. The additional cost may be recovered from the gain in electrical power achieved by the cooled PV through PCM depending on site climatic conditions, design of PV-PCM system and selection of the PCM (Browne et al., 2015; Hasanuzzaman et al., 2016).

Although the PV-PCM systems are studied extensively as reviewed by (Browne et al., 2015; Ma et al., 2015; Islam et al., 2016), a careful analysis of previous studies (summarized in Table 2) reveals that all the experimental studies were conducted for few days only. A year long experiment is deemed indispensable in order to understand the PCM deterioration over repeated thermal cycles when subjected to extremely high temperature climates (Hasan et al., 2014b). The Previous studies suggested that PCMs phase transition temperature needs to be optimized based on seasonal variation, orientation and location of PCM layer (Kibria et al., 2016). Previous findings recommended that the night time

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Nomenclature

A	area (m^2)
c	heat capacity (kJ/kg K)
c_o	solid heat capacity (kJ/kg K)
FF	fill factor
G	solar radiation intensity (W/m^2)
h	heat losses coefficient (W/m^2K)
H	convection matric
I_{sc}	short circuit current (A)
k	thermal conductivity (W/m K)
K	conductivity matrix
L	latent heat of fusion (kJ/kg)
M	mass matrix
PCM	phase change material
P_e	electric power (W)
P_s	energy savings (W h)
PV	photovoltaics
$\hat{q}$	irradiance or boundary flux matrix
R	radiation matrix
t	time (min)
T	reference temperature (°C)
T_{amb}	ambient temperature (°C)
$T_{amb\text{-}avg\text{-}day}$	average daytime ambient temperature (°C)
$T_{amb\text{-}avg\text{-}nig}$	average nighttime ambient temperature (°C)
T_l	liquidus temperature (°C)
T_{PV}	PV panel temperature (°C)
T_s	solidus temperature (°C)
$\dot{T}$	time derivative of temperature
V_{oc}	open circuit voltage (V)
V_W	wind speed (m/s)
x_i and x_j	unit vectors
Z_c	convection heat transfer (W h)
Z_r	radiation heat transfer (W h)
ρ	density (kg/m^3)
η	energy efficiency

ambient temperatures dictate PCM regeneration to solid. The PCM phase transition temperature should be higher than the average night time summer temperature to ensure PCM regeneration to solid in the worst case i.e., warmer nights (Hasan et al., 2014a). Additionally, the PV panel temperature (T_{PV}) during day time in winter should be substantially higher than the PCM liquidus temperature (T_l) in order to ensure complete PCM melting in colder months for effective heat removal.

Table 1
Advantages and disadvantages of different thermal management techniques.

	Advantages	Disadvantages
Natural air circulation	– low initial cost – no maintenance – easy to integrate – longer life – no noise – no electricity consumption – passive heat exchange	– low heat transfer rates – accumulation of dust in inlet grating further reducing heat transfer – dependent on wind direction and speed – low thermal conductivity and heat capacity of air – low mass flow rates of air – limited temperature reduction
Forced air circulation	– higher heat transfer rates compared to natural circulation of air – independent of wind direction and speed – higher mass flow rates than natural air circulation achieving high heat transfer rates – higher temperature reduction compared to natural air circulation	– high initial cost for fans and ducts to handle large mass flow rates – high electrical consumption – maintenance cost – noisy system – difficult to integrate compared to natural air circulation system
Hydraulic cooling	– higher heat transfer rate compared to natural and forced circulation of air – higher mass flow rate compared to natural and forced circulation of air – higher thermal conductivity and heat capacity of water compared to air – higher temperature reduction	– higher initial cost due to pumps – higher maintenance cost compared to forced air circulation – higher electricity consumption compared to forced air circulation – less life compared to forced air circulation due to corrosion
Heat pipes	– passive heat exchange – low cost – easy to integrated	– low heat transfer rate – dust accumulation on the inlet grating – dependent on the wind speed and direction
Thermoelectric (Peltier) cooling	– no moving parts – noise Free – small size – easy to integrate – low maintenance costs – solid state heat transfer	– heat transfer depends on ambient conditions – active systems – require electricity – reliability issues-costly for PV cooling – no heat storage capacity – requires efficient heat removal from warmer side for effective cooling
PCM thermal management	– higher heat transfer rate compared to both forced air circulation and forced water circulation – higher heat absorption due to latent heating – isothermal nature of heat removal – no electricity consumption – passive heat exchange – no noise – no maintenance cost – on demand heat delivery	– higher PCM cost – some PCMs are toxic – some PCMs have fire safety issues – some PCMs are strongly corrosive – PCMs may have disposal problem after their life cycle is complete

Table 2
Summary of findings of experimental research on PV cooling using PCM under various outdoor conditions.

Ref.	Methodology	Weather condition	PCM tested	Period	Findings
Hasan et al. (2015)	Experimental numerical	Cool warm	Salt hydrate fatty acids	14–16 days	The daily average power savings was found to be 7.7% in hot climate in the best case
Hassan et al. (2014)	Experimental numerical	Hot	Paraffin, Salt hydrate, fatty acids	12 hrs	The maximum drop in peak temperature by using PCM was 5 °C on a cloudy day and 11 °C on clear sky condition
Hasan et al. (2014b)	Experimental numerical	Cool warm	Fatty acids Salt hydrate	14–16 days	System is cost effective only in warmer climates
Aelenei et al. (2014)	Experimental numerical	Cool	–	2-days	The maximum electrical efficiency of this BIPV-PCM can reach 10% and the thermal one 12%
Biwole et al. (2013)	Experimental numerical	–	Acronym SP/PCM paraffin		A time lag of 1 h was observed to reach 34.9 °C while the PCM always maintained PV temperature below 50 °C
Hendricks and Sark (2013)	Experimental numerical	Moderate hot		Year	Energy gain from PV-PCM system was 3.3 kW h in hot and 1.8 kW h in cold climate
Huang (2011)	Experimental numerical	Cool	Combinations of paraffin	3-days	Combination of RT27 and RT21 achieves the highest temperature reduction during the daily operation
Sharma et al. (2016)	Experimental	–	Paraffin	160 min	The open-circuit voltage improvement, at 1000 W/m^2 was 4.4% with a reduction in module temperature of 3.8 °C
Sarwar (2012)	Experimental numerical	–	Lauric acid palmitic acid	–	Lauric acid and palmitic acid PCMs were found to drop the module temperature by 22 °C and 19.5 °C respectively
Maiti et al. (2011)	Experimental numerical	Indoor outdoor	Paraffin wax metal-wax composite	7 hrs	The PV-PCM system with embedded metal shreds maintains the temperature around 65–68 °C for 3-h compared with paraffin only which reached 84 °C in less than 50 min
Lillo et al. (2011)	Experimental	12-sites	PCMs with different melting points	–	CPV-PCM system yields 37.2% more power than CPV-only system in hot climate and 18% in cold climate
Wu (2009)	Experimental	–	Paraffin	–	A temperature drop of 18 °C with 10% increase in electrical efficiency is achieved by the CPV-PCM system at high intensity 672 W/m^2

Employing Al Ain weather data from climate consultant (Climate consultant, 2016), average night time temperature ($T_{amb\text{-}avg\text{-}nig}$) in summer (June) was found to be 34 °C. The average PV temperature ($T_{PV,avg}$) during day time in winter (December) was found to be ≈46 °C calculated by employing Eq. (1) developed by (Skoplaski and Palyvos, 2009). Based on the criteria developed by the authors, A PCM with T_s higher than 34 °C and T_l lower than 46 °C would be an optimum choice.

$$T_p = T_{a+}\left(\frac{0.32}{8.91 + 2V_w}\right)G \tag{1}$$

The comparison of thermo-physical, kinetic and chemical properties of PCMs from salt hydrates, paraffin waxes and fatty acids (Hasan et al., 2014a) concluded that:

- The paraffin waxes possess thermal conductivity (0.2 W/m K), higher than that of fatty acids (0.14 W/m K) and lower than that of salt hydrates (1.08 W/m K).
- The paraffin waxes have heat of fusion (139 kJ/kg) comparable to salt hydrates (210 kJ/kg) and fatty acids (168 kJ/kg).
- The paraffin waxes have density (0.88 kg/m^3) comparable to that of fatty acids (0.89 kg/m^3) however, lower than that of salt hydrates (1.7 kg/m^3).
- The paraffin waxes show no sub-cooling while the salt hydrates and fatty acids show a higher and a moderate sub-cooling respectively.
- The paraffin waxes are not corrosive to metals while the salt hydrates and fatty acids do show corrosion to the metals (Hasan et al., 2014a).

Since most of the favorable characteristics are found in paraffin waxes, a paraffin based PCM with T_s of 38 °C and T_l of 43 °C namely RT40 (Rubitherm, 2016) is selected for the current experiment.

2. Methodology

The research methodology consists of identifying an optimum PCM melting point with the help of the historical weather data, selecting a suitable PCM type with the desired properties outlined in Table 3 and an experimental evaluation of the PCM performance for cooling of the PV for the whole year in the hot climate of UAE. The PV cooling and associated improvement in the PV power is quantified. The cooling is quantified by comparing the temperature

Table 3
Properties of a PCM desired for photovoltaic thermal regulation.

	Requirement	Reason for requirement
Properties		
Thermal	High latent heat	Maximum heat absorption
	High heat capacity	Minimum sensible heating
	Good thermal conductivity	Efficient heat removal
	Reversible phase change	Diurnal response
	Fixed melting point	Consistent behavior
Physical	Congruent melting	Minimum thermal gradient
	Low volume expansion	No overdesign
	High density	Low containment requirement
Kinetic	No supercooling	Easy to freeze
	Good crystallization rate	Faster solidification
Chemical	Chemical stability	Long life
	Non-corrosive	Long container life
	Non-flammable	Comply building safety codes
	Non explosive	Environment friendly
	Non-toxic	–
Economic	Abundant	Market competitiveness
	Cheap and cost effective	Economic viability and market penetration
Environmental	Recyclable/reusable	Ease to dispose off
	Odour free	Comfortable to apply in dwellings environment

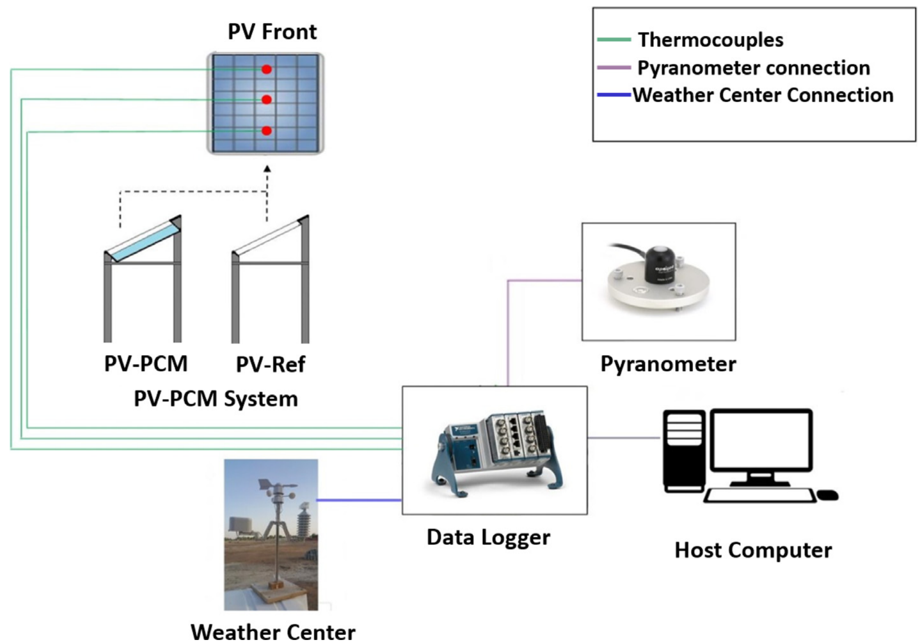

Fig. 1. Experimental setup consisting of PV-reference and PV-PCM deployed outdoors in Al Ain, United Arab Emirates and connected to data acquisition system.

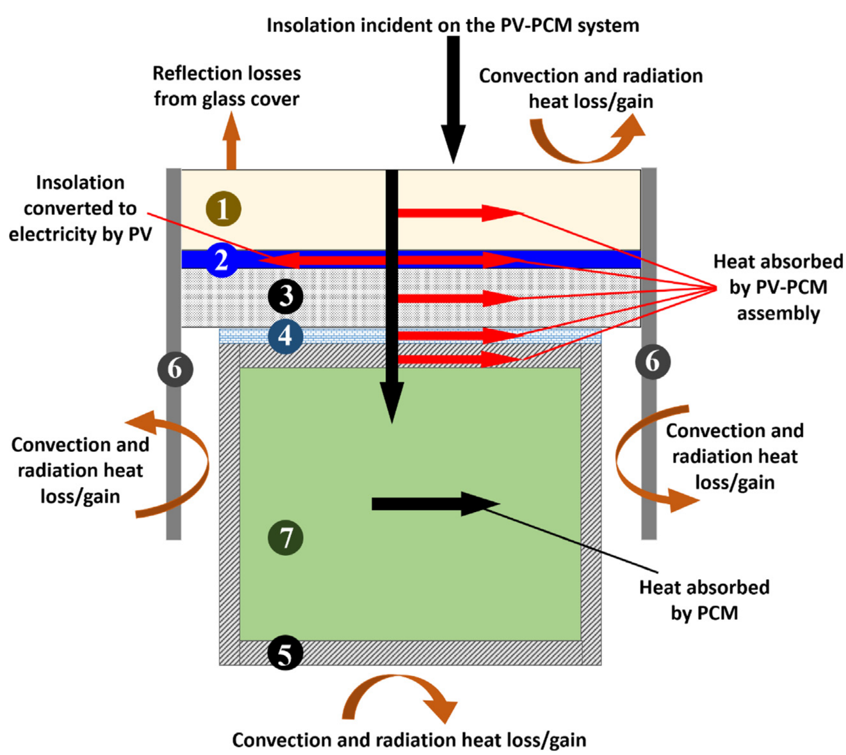

Fig. 2. The schematics of the energy flow in the PV–PCM system under investigation with labels 1 – glass cover, 2 – PV cell, 3 – PV back sheet, 4 – layer of epoxy glue as interface between PV and PCM container, 5 – PCM container wall, 6 – PV frame and 7 – PCM layer.

at the PV front surface achieved with and without PCM. The enhanced PV power is calculated by employing fill factor (FF) on measured open circuit voltage (V_{oc}) and short circuit current (I_{sc}) of the respective PVs. A thermal model is developed and validated with the experimental results. The validated model is employed to predict PCM melt and solidification fractions in each month of the year. The melt/solidification fraction indicates progress in PCM melting/solidification and effect of the same is observed on the PV cooling in different seasons of the year.

2.1. Experimental setup

Polycrystalline EVA-encapsulated PV panels with dimensions of 53 cm × 63 cm, rated capacity of 40 W and module efficiency of 15% were tested outdoors facing south at the latitude of Al Ain (24.1° N, 55.8° E). The electrical performances of the modules were confirmed by measuring the V_{oc} and I_{sc} prior to inclusion of the PCM in the panels. One PV panel was treated as a reference and the other was glued by epoxy resin to internally finned metallic

Table 4
Thermo-physical properties of PCM, metallic and bonding materials used in the experiments in solid state.

Properties	Melting point (°C)	Congealing point (°C)	Latent heat (kJ/kg)	Specific heat capacity (kJ/kg K)	Heat conductivity (W/m K)	Density (kg/L)	Volume expansion	Flash point (°C)
PCM RT42 (a)	38–43	43–37	145 ± 7.5%	2	0.2	0.88	12.5%	186
Polystyrene (b)	240	NA	NA	NA	0.032	NA	NA	350
Aluminum (c)	650	NA	NA	0.91	222	2.71	24×10^{-6}/K	NA
Epoxy Resin (d)	130	NA	NA	NA	1.26	2.09	34×10^{-6}/K	350

a: Rubitherm (2016).
b: International Program on Chemical Safety (2016).
c: Aalco Metals Ltd (2016).
d: Electrolube (2016).

containers filled with the solid PCM. The containers and internal fins were fabricated from a 4-mm thick sheet of aluminum alloy (1050A).

Multiple T-type copper-constantan thermocouples were installed on the two PV systems as shown in Fig. 1. The thermocouples were attached to the front and back PV surfaces. The thermocouples positioned on the PV front surface were shielded from direct irradiation by the opaque insulation tape. A self-powered pyranometer calibrated at 0.20 mV/(W m^{-2}) (Apogee Instruments, 2016) was installed at the latitude angle of the site (24.1° N, 55.8° E) to measure solar radiation intensity (G). A weather station (WS1041, 2016) was installed to measure the ambient temperature (T_{amb}) and wind speed (V_w). All the sensors were connected to a data logger (National Instruments-NI, Compact-Rio) to store and retrieve the data.

2.2. Experimental procedures

The PCM was filled as liquid (10.2 L) in the containers attached to the PV and subsequently cooled until it was completely solidified. The solidified PCM left a 7-cm free space on top of the container which was intended to accommodate volume expansion during the PCM melting. The experiments were conducted for one year from 01/01/2015 to 31/12/2015. The PV panels data of the T_{PV}, V_{oc} and I_{sc} were measured for the reference PV and for the PV integrated with the PCM (PV-PCM) simultaneously with weather data of G, V_w and T_{amb} with time step of 5 min for the whole year. Although occasionally rain, storm and haze intervened with data logging, the weather generally remained stable to execute data logging for at least 20 days each month. The results are, however, presented for a typical selected day of each month to avoid repetition. The net yearly energy savings are computed by aggregating the PV performance data for all the available days.

3. Numerical modelling

A two-dimensional heat transfer model is developed and applied to simulate the cooling of PV by the PCM conceptually illustrated in Fig. 2 (Sarwar, 2012; Hasan et al., 2015). The differential Eq. (2) governs the transient heat transfer in two dimensions (Reddy and Gartling, 2001). Heat flux (q) representing incident irradiance is applied as boundary condition at the PV surface.

$$\rho c\frac{\partial T}{\partial t} - \left[\frac{\partial}{\partial x_i}\left(k_{ij}\frac{\partial T}{\partial x_j}\right)\right] = 0 \tag{2}$$

where ρ is density, c is heat capacity, k is thermal conductivity, T is temperature, t is time and x_i, and x_j are unit vectors. Eq. (2) accounts for conduction in the domain ignoring boundary heat losses. The heat losses (convection and radiation) are imposed at the boundary as given in Eq. (3) and (4):

$$Z_c = h_c A(T - T_{amb}) \tag{3}$$

$$Z_r = \sigma \varepsilon A\left(T^4 - T_\infty^4\right) \tag{4}$$

Z_c and Z_r are convective and radiative heat losses, h_c is convective heat loss coefficient, A is PV area, ε is PV emissivity, σ is Stefan-Boltzman constant, T_{amb} is ambient temperature and T_∞ is the sky temperature. The unified governing heat transfer equation imposing heat losses yields Eq. (5).

$$\rho c\frac{\partial T}{\partial t} - \left[\frac{\partial}{\partial x_i}\left(k_{ij}\frac{\partial T}{\partial x_j}\right)\right] + Z_c + Z_r = 0 \tag{5}$$

The weak formulation is generated by multiplying test function δT and integrating over the domain resulting in the Eq. (6):

$$\underbrace{\int_\Omega \delta T \cdot \rho c\frac{\partial T}{\partial t}\partial\Omega}_{1} - \underbrace{\int_\Omega \delta T \cdot \left[\frac{\partial}{\partial X_i}\left(k_{ij}\frac{\partial T}{\partial X_j}\right)\right]\partial\Omega}_{2} + \underbrace{\int_{\Gamma_\sigma} \delta T \cdot (z_c + z_r)\partial A = 0}_{3} \tag{6}$$

The weak formulation is simplified by applying Green-Gauss and divergence theorem on part 2 of Eq. (6) resulting in Eq. (7):

$$\underbrace{\int_\Omega \delta T \cdot \rho c\frac{\partial T}{\partial t}\partial\Omega}_{1} + \underbrace{\underbrace{\int_\Omega \left[k_{11}\frac{\partial \delta T}{\partial X_1}\left(\frac{\partial T}{\partial X_1}\right) + k_{22}\frac{\partial \delta T}{\partial X_2}\left(\frac{\partial T}{\partial X_2}\right)\right]\partial\Omega}_{i} - \underbrace{\int_\Gamma \delta T.\hat{q}\partial A}_{ii}}_{2} \cdots\cdots\cdots + \underbrace{\int_\Gamma \delta T.(z_c + z_r)\partial A}_{3} = 0 \tag{7}$$

The energy balance is retained in weak formulation for 2D transient differential heat diffusion given in Eq. (8).

$$M\dot{T} + KT - \hat{q} + (H + R)T = 0 \tag{8}$$

where $\dot{T}$ is time derivative of temperature, $\hat{q}$ is the irradiance or boundary flux matrix and M, K, H and R are the mass, conductivity, convection and radiation matrices respectively. The temporal discretization of the domain is carried out by Crank-Nicholson method (Reese, 2003 and Kwon and Bang, 1997). The heat storage during phase change is simulated by applying effective heat capacity method (Lamberg et al., 2004) given in Eq. (9):

$$c_{p,e} = c_0 + \frac{L}{T_s - T_l} \; if \; T_s \leqslant T \leqslant T_l \; or \; else \; c_{p,e} = c_0 \tag{9}$$

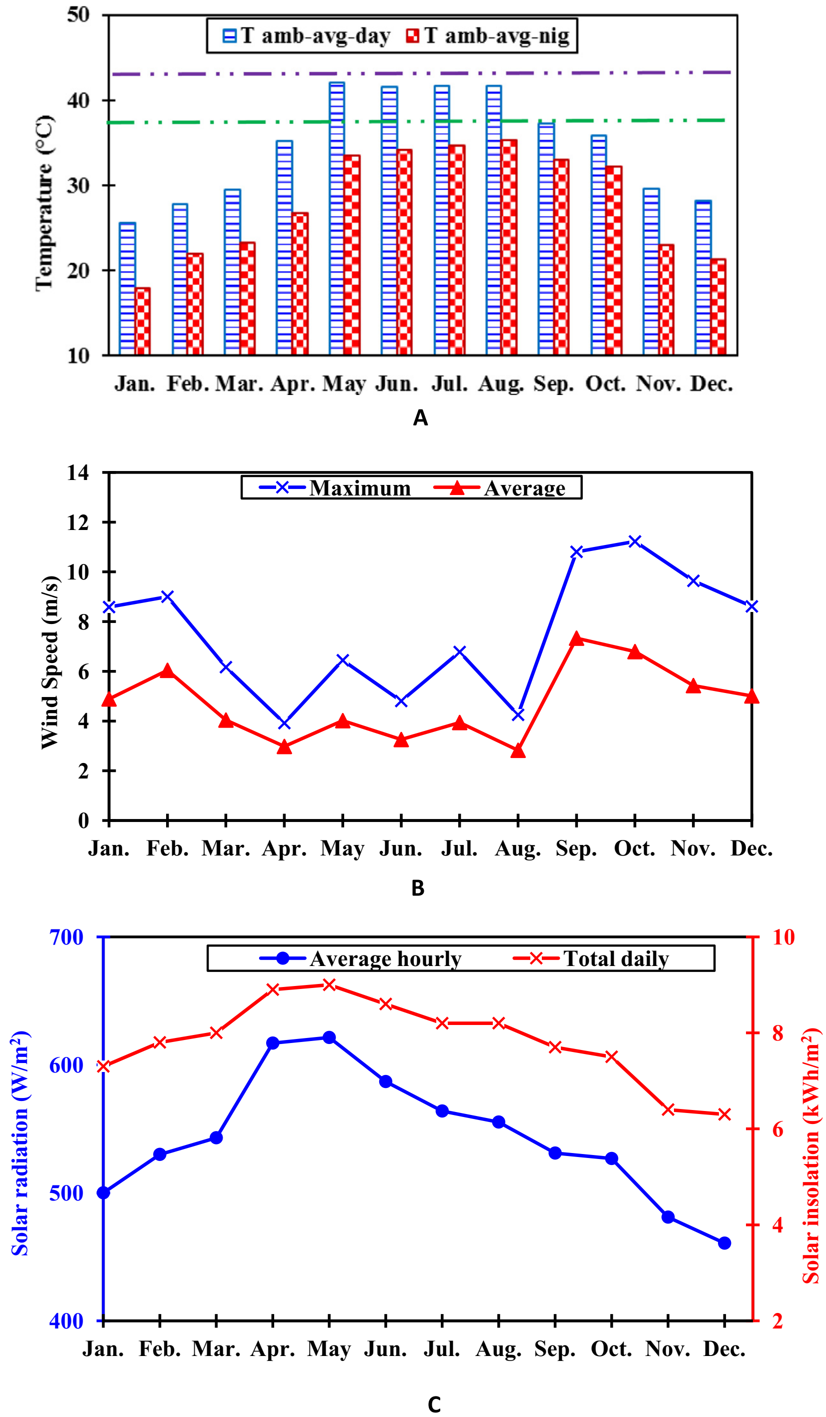

Fig. 3. Weather data for a typical day of each month represented by average day and night time ambient temperature, average hourly and total solar radiation and average and highest wind speed.

where c_o is the reference heat capacity (solid), L is the latent heat of fusion, T_s is solidus temperature and T_l is liquidus temperature.

The developed model is employed to predict the transient temperature distribution within a two-dimensional region in the PV-PCM system for different G, T_{amb} and convective and radiative heat transfer boundary conditions. A series of simulations at various ambient conditions in different seasons of the year have been carried out employing the material properties listed in Table 4. The results of the year-long experiment are compared with the prediction of the simulation model taking one sample day of each month. The comparison of results obtained for the PV panel only and PV–PCM are discussed in Section 4.

4. Results and discussion

The weather data represented by T_{amb}, G and V_w are the most important parameters which influence the PV heat gain/loss thereby dictating the PV panel temperature (T_{PV}). The changing weather conditions result in different T_{PV} in different seasons of the year. The variation in T_{PV} eventually affects the PCM performance as a temperature regulator as well. Generally, the climate of UAE is hot and humid with prolonged summer and a modest winter. The peak summer spans over May-Aug. reaching peak temperature of 49 °C with an average temperature of 34–39 °C. The moderate winter spans from December to January having lowest temperature of 14 °C with a range of 22–27 °C. Fig. 3(A–C) presents the on-site measured weather data for an average day of each month in Al-Ain, UAE.

Fig. 3A shows the average ambient temperature during a day ($T_{amb\text{-}avg\text{-}day}$) and a night ($T_{amb\text{-}avg\text{-}nig}$) for a typical diurnal cycle of each month of the year 2015. Average ambient temperature ($T_{amb\text{-}avg}$) is an effective way to assess PCM melting/solidification performance due to free heating/cooling during day/night. It can be seen that $T_{amb\text{-}avg\text{-}day}$ remained lower than T_l (43 °C) indicating that the ambient temperature alone in the absence of solar radiation (cloudy day) would not have completely melted the PCM in any month of the year. $T_{amb\text{-}avg\text{-}nig}$ dictates the PCM solidification at night time due to passive heat dissipation to ambient. From Fig. 3A $T_{night\text{-}avg}$ remained lower than T_s of the PCM (38 °C) for the whole year indicating that ambient air circulation (natural or forced) at night would be effective to cool the PCM at night and regenerate it to solid for the next day.

Fig. 3B shows the average and the peak V_w which determines the convection heat loss/gain from the PV surface. It can be seen that the average V_w varied between 3 m/s and 7 m/s in different months, however, it generally remained lower during summer months (3.2 m/s). The average h calculated employing the Eq. (10) introduced by (Sharples and Charlesworth, 1998) ranged between 15.7 W/m^2 K and 23.5 W/m^2 K in different months. The peak V_w exhibited similar trend, however, with a higher magnitude ranging between 4 m/s and 11.5 m/s. The peak h values ranged between 18.3 W/m^2 K and 37.8 W/m^2 K in different months of the year. Lower h along with higher T_{amb} in summer months could lead to the reduced PCM heat loss thereby affecting its solidification at night time.

$$h = 7.9 + 2.6V_w \tag{10}$$

Fig. 3C shows the average and peak G and the total daily insolation which determine the energy incident on the PV eventually controlling T_{PV}. The total solar radiation received by the PV is potentially important to determine the amount of the PCM required to size the PCM containment. G along with total daily insolation was higher during the summer months (April–July) and lower during the winter months (November–January). The total daily insolation ranged from 6.5 kW h/m^2 to 9 kW h/m^2 between the winter and the summer showing a variation of ~38%. It indicates a potential problem of over-sizing/under-sizing of the PCM containment by same amount if based on highest/lowest daily solar radiation received by the PV.

Fig. 4 shows experimental and simulated transient temperature evolution on the front surface of PV panel for a typical day in January (the winter month) and July (the summer month). It can be seen that the peak T_{PV} reached 53 °C in January and 72 °C in July indicating a variation of 36%, similar to the variation observed in total daily insolation (38%). The variation in peak T_{PV} affects the PCM performance as the PV temperature regulator, since heat extracted by the PCM depends on the difference between the T_{PV} and the T_s. In order to extract an optimal amount of heat from the PV, a similar shift in the T_s as of the T_{PV} would be desired which is not possible as a PCM can only have one fixed T_s. The varying T_{PV} in different seasons, therefore, affects PCM temperature regulation performance emphasizing the need for the right selection of T_s for an optimal year round performance. The peak reference T_{PV} was higher than the T_l (43 °C) in all months which indicates that the PCM always could melt by absorbing heat from the PV and regulate its temperature. It can be seen that the PV-PCM rendered a temperature lag behind reference PV in all the months indicative of cooling effect produced by the PCM. The experimental and simulated temperatures agreed indicating the accuracy of the developed thermal model. Similar trends in experimental and simulated temperatures are observed for rest of the months presented in Appendix A.

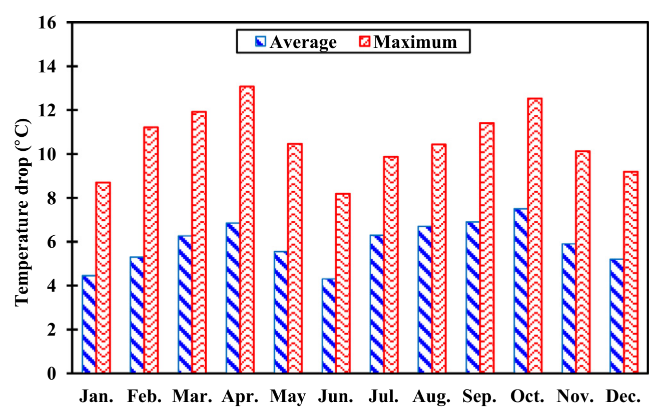

Fig. 5. Average and peak temperature drop achieved by PV-PCM for a typical day of each month.

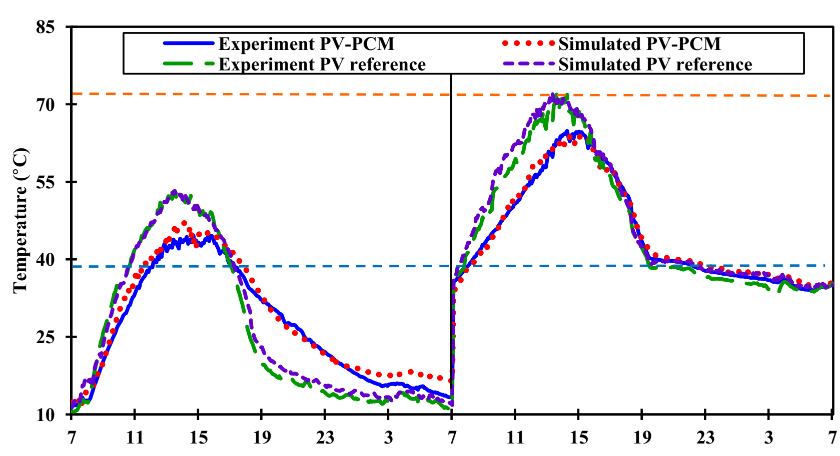

Fig. 4. The experimental and simulated transient temperature rise of the PV and PV-PCM system for a typical day of the coolest (January) and warmest (July) months.

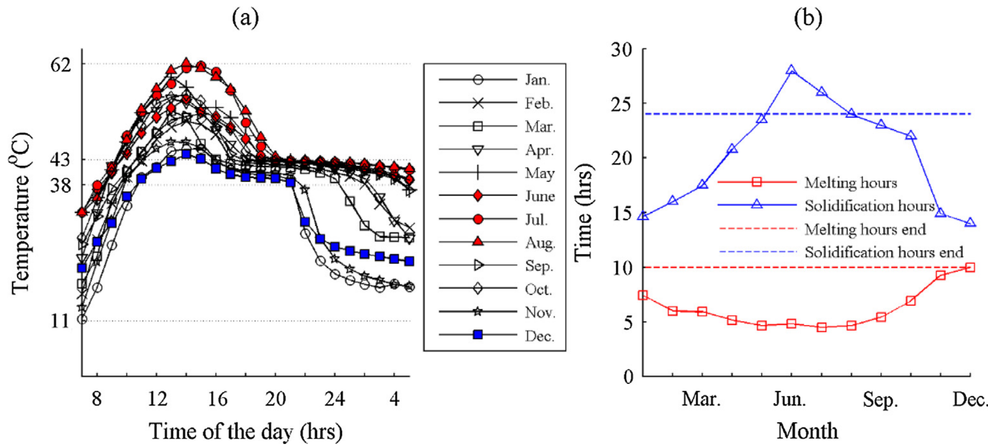

Fig. 6. (a) Peak temperature variation of the PCM. (b) Time taken by the PCM for complete melting and solidification in a typical day of each month.

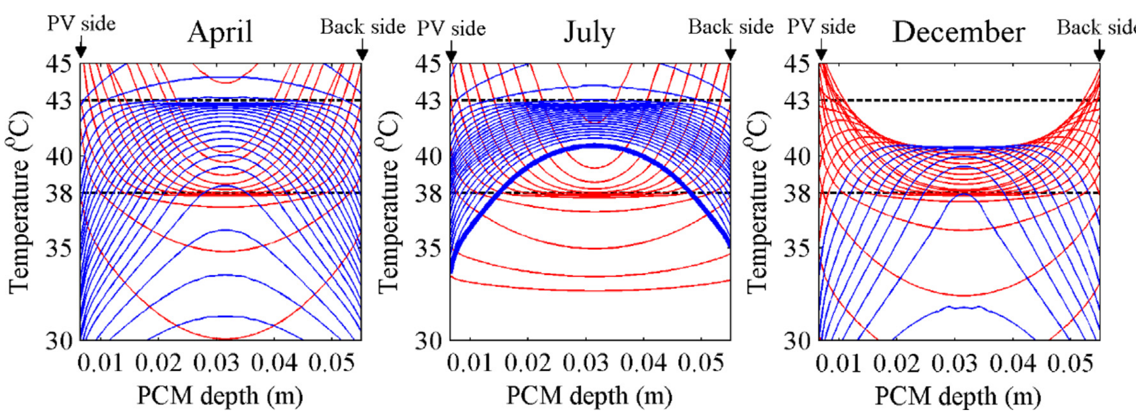

Fig. 7. Temperature curves in the PCM for every 30 min during melting (red curves) and solidification (blue curves) for a typical day in the month of April, July and December. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 5 presents the average and peak temperature drops on PV front surface achieved by inclusion of the PCM. It can be seen that the winter and summer months produced less temperature drop compared to moderate months. The peak and average temperature drops of 13 °C and 6 °C respectively, were exhibited in April which dropped to 8.6 °C and 2.3 °C respectively in January. The lower temperature drop (average and peak) observed from June–August can be explained by incomplete solidification of the PCM in warmer nights of the summer months presented in Fig. 6. Inclusion of the PCM achieved a 10.5 °C peak time decrement in the T_{PV} as an average on the yearly basis. Since the long peak hours contribute to the major share in electricity production in PV, peak time temperature drop (by 10.5 °C in the present case) can achieve concomitant substantial energy savings.

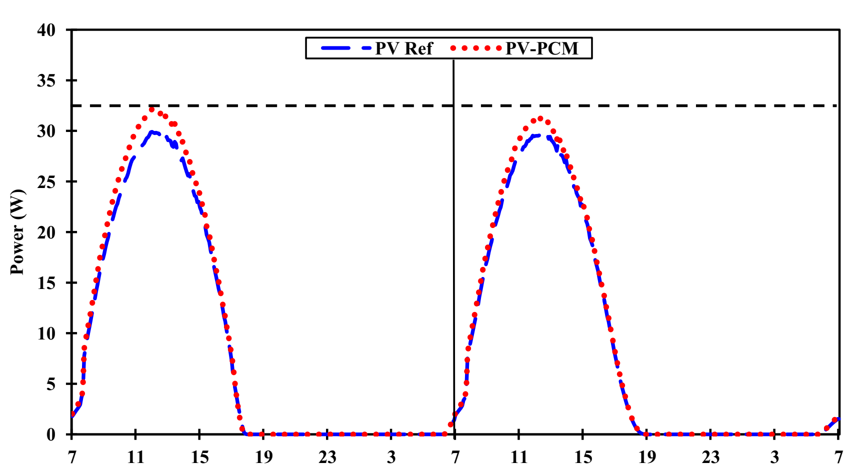

Fig. 8. The transient power production from the reference PV and the PV-PCM system for a typical day of the coolest and the hottest month.

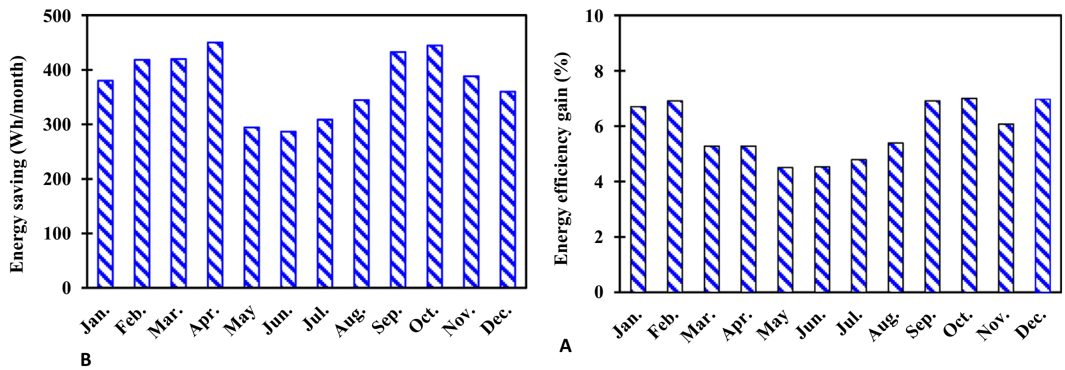

Fig. 9. Monthly energy savings (A) and energy efficiency enhancement (B) achieved by PV-PCM through the PV cooling.

Table 5
Accuracies and measurement ranges of the devices used in experiments for weather data, temperature and PV performance measurements and data acquisition.

Measurement parameter	Device	Model	Measurement range	Measurement error
Solar radiation (Apogee Instruments, 2016)	Apogee pyranometer	SP-110	NA	±1%
Current (National Instruments, 2016)	NI-Analogue module	9227	5 Arms	±0.01%
Surface temperature logging (National Instruments, 2016)	Ni-Analogue module	9213	−75 to 250 °C	±1%
Surface temperature measurement	Thermocouples	t-type		±1%
Ambient temperature (WS1041 Professional Weather Station, 2016)	Starmeter weather station	WS1041	−40 to 60 °C	±1%
Wind speed (WS1041 Professional Weather Station, 2016)	Starmeter weather station	WS1041	0–50 m/s	±1%
Voltage (National Instruments, 2016)	NI-Analogue module	9221	−60 to 60 V	±0.25%

Fig. 6a shows the temperature of the PCM over time while Fig. 6b shows the time taken by the PCM to completely melt and solidify predicted by the validated model. The PCM melted completely in shorter time in summer months, however, it took longer time to solidify at night due to higher T_{amb}. The opposite was observed in the winter months. As shown in Fig. 6a, the inclusion of PCM resulted in a time shifting of the peak PV temperature. The peak temperature of the PCM was achieved between 13:00 and 14:00 local time over the whole year. The gradient of temperature curve decreased during phase transformation due to the latent energy storage and release respectively. As soon as phase transition completed a sharp change in gradient of PCM temperature curve was observed due to sensible heating/cooling. The melting time was assumed to be from 7:00 to 17:00 (10 h) as shown by the red dotted line in Fig. 6b while the solidification time was assumed to be from 17:00 to 7:00 next morning (14 h,) as shown by the blue dotted line in Fig. 6b. It shows that the PCM completely melted in 4 h in the peak summer months (June–August) while it took more than 10 h to melt in a peak winter month (December). Furthermore, the PCM did not completely solidify in the peak summer months of June, July and August as shown in Fig. 6a and b due to the higher T_{amb} represented by the solid fraction in the range of 0.71–0.89. Such a huge variation in the melting time renders constraint on the sizing of the PCM containment and would need to be optimized. To understand the melting and solidification behavior of the PCM, the temperature curves of the PCM over its thickness were plotted every 30 min for three selected months (April, July and December) shown in Fig. 7. These months are selected to show three different behaviors of the PCM which are:

- The PCM completely melts and solidifies in April
- The PCM completely melts but does not completely solidify in July
- The PCM completely solidifies but does not completely melt in December

As shown in Fig. 7, the red temperature curves show charging of the PCM (when the PCM absorbs heat) while blue temperature curves show discharging of the PCM (when the PCM releases heat). The results for winter (December) show that the PCM melts near to the PV side and the back side walls but could not completely melt in the center of the container reaching a melt fraction of 0.89 at the end of the day. Incomplete melting consequently rendered less temperature drop by the PCM in December. The temperature curves for the month of July shows that the PCM could not completely solidify in the center of the container. The lowest temperature of ~40.5 °C was found in the center of the container at the end of the night as shown by the bold blue line in Fig. 7 reaching solid fraction of 0.78. Incomplete solidification affected the PCM performance in the peak summer months which consequently yielded a less temperature drop in the PV (Fig. 5 and Appendix A). The moderate months (April and October) showed a complete melting and solidification as shown in Fig. 7 (for April) thereby achieving higher temperature drop (Fig. 5). The temperature curves presented in Fig. 7 also show that the PCM melts and solidifies completely near to PV side and back side walls over the whole year. The PCM could not either completely melt in the peak winter or completely solidify in the peak summer in the middle of the container. This observation provides an insight to design a system that allows a complete melting and solidification over the whole year. One option could be to start air/water circulation at early hours in the summer months to extract heat from the melted PCM in order to allow extended time for PCM to solidify. The other option could be to introduce several layers of PCMs in the container having different phase transformation temperatures to ensure a complete melting and solidification over the whole year.

The drop in T_{PV} resulted in increased transient PV power output (P_e) presented in Fig. 8 for a typical day of the above mentioned hottest and coldest months. The P_e is calculated from measured V_{oc} and I_{sc} applying FF given by Eq. (11).

$$P_e = \frac{V_{oc} I_{sc}}{FF} \tag{11}$$

The FF was deduced from the PV catalogue data being 0.72. The reference PV and PV-PCM exhibited similar V_{oc} and I_{sc} profiles, however the PV-PCM produced higher P_e compared to the reference PV which was more apparent at the peak time. The rest of the months showed similar profile however with different magnitudes of P_e presented in Appendix B. The monthly power saving (P_s) and energy efficiency enhancement (η) achieved by PV-PCM by virtue of temperature drop are presented in Fig. 9A and B, respectively. The P_s is calculated by subtracting the P_e produced by the reference from the P_e produced by PV-PCM given by Eq. (12).

$$P_s = P_{e,PV-PCM} - P_{e,ref} = \left(\frac{V_{OC} I_{SC}}{FF}\right)_{PV-PCM} - \left(\frac{V_{OC} I_{SC}}{FF}\right)_{ref} \tag{12}$$

The η is calculated by dividing the P_s by the P_e produced by reference PV given in Eq. (13).

$$\eta = \frac{P_s}{P_{e,ref}} \tag{13}$$

As expected, the warmest month showed lowest while the moderate months showed the highest P_s as well as η, similar to the temperature drop observed in Fig. 4. The PV cooling by the PCM achieved an average power saving of 5.9% on a yearly basis.

5. Economic analysis

Integrating Eq. (12) over time, the amount of extra electricity produced was 13.5 kWh/m^2 which when multiplied by the rate of the international electricity (Germany at 0.15 Euro/kWh) brings the economic benefit of 2.2 dollars/m^2. The additional cost incurred due to inclusion of the PCM at the rate of 1 dollar/kg for paraffin wax (Alibaba, 2016) incurred a payback period of 10 years. The relatively higher payback period is caused by higher paraffin material cost compared to salt hydrates priced at 0.14–0.24 dollar/kg (Alibaba, 2016). Since the paraffin has similar heat of fusion as salt hydrates, employing the same amount of the salt hydrate would yield same electrical energy gain. When recalculated with salt hydrate as PCM, the payback fell down to 3 years. The PCM cost is expected to drop with increased interest in PV-PCM systems thus creating an optimal economic scenario (Browne et al., 2015).

6. Experimental uncertainties

The uncertainties involved in the measured parameters are predicted from the errors of the measuring devices listed in Table 5. The uncertainty in solar radiation and wind speed are found to be 1% while that of voltage, current, and temperature are found to be 0.25%, 0.01% and 2% respectively. The cumulative uncertainty involved in the measured data reached up to 5.26%. This uncertainty is around the expected range of 5% confirming the accuracy of the experimental setup. Employing experimental uncertainty on the measured data, the total yearly energy saving varied by ±0.003%.

7. Conclusions

The PV-PCM system employing an optimum melting point paraffin based PCM is studied in the hot weather conditions of Al Ain, UAE. It is observed that the PCM shows variant performance in different months of the year. The PCM achieved the peak-time temperature drop of 13 °C in April and 8 °C in June. Accordingly, PCM achieved higher average temperature drop of 6 °C in April compared to that of 2.3 °C in June. An experimentally validated model predicted the PCM melting and solidification times and melt fractions to help understand the deviation in PCM cooling performance in different months. In summer, the PCM could not completely solidify due to warmer ambient temperature during the night that hampered its performance. While in winter, the PCM could not receive enough thermal energy to completely melt eventually reducing its cooling performance. The maximum cooling was observed for the moderate months in April and October. In moderate months the PCM completely melted during day time due to enough irradiance and ambient temperature and solidified at night time due to lower enough ambient temperature. The PCM eventually achieved 10.5 °C drop in PV temperature on average at peak time resulting in 5.9% increase in PV power output on yearly basis.

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Appendix A

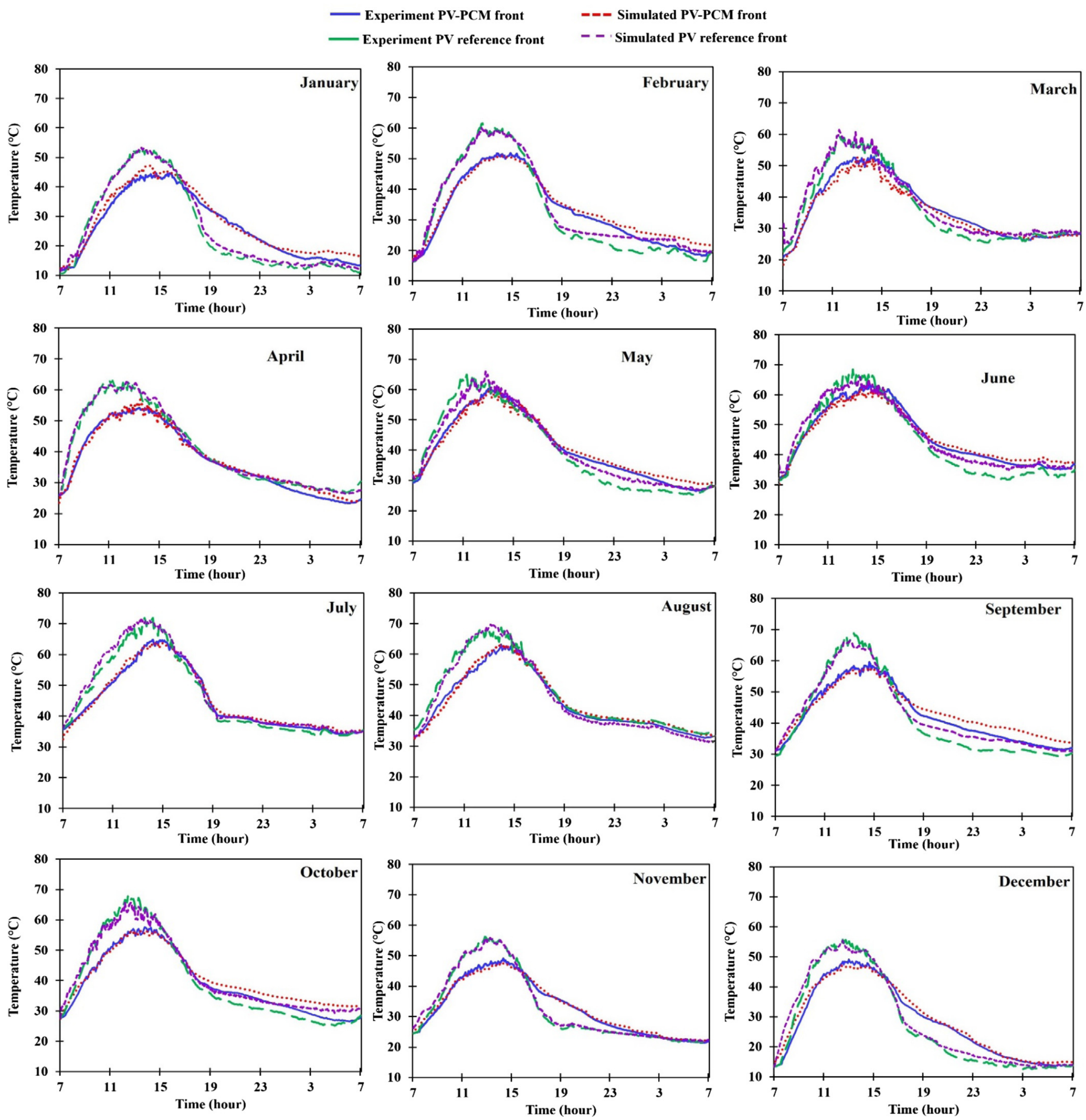

Fig. A. The experimental and simulated transient temperature rise of the PV and PV-PCM system for one typical day of each month of the year.

Appendix B

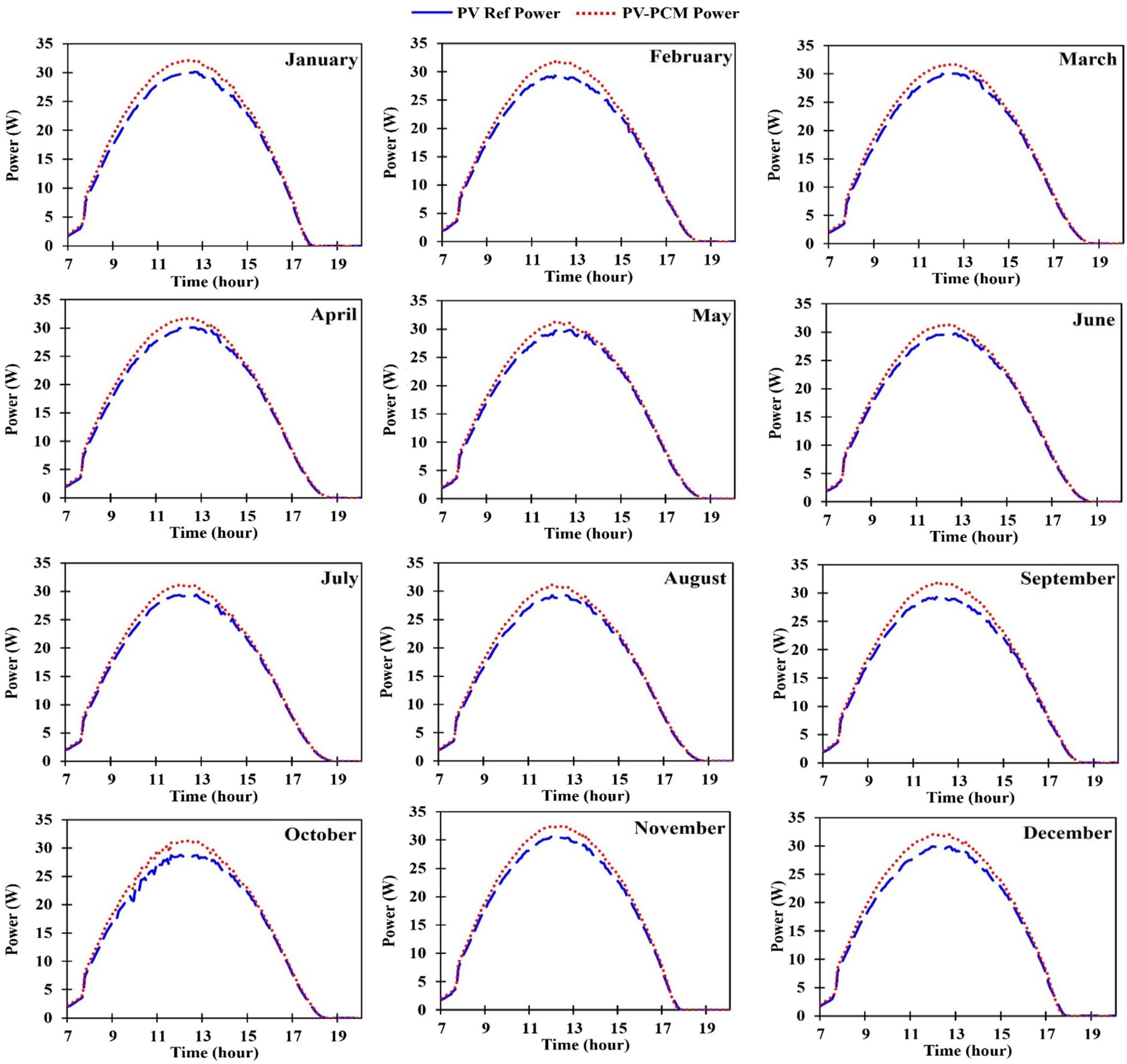

Fig. B. The transient power production from the reference PV and the PV-PCM system for a typical day of each month of the year.

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